GenomeAxis Reference Laboratory

Comprehensive Tumor Profiling — Pediatric Brain Panel v3.2

Patient	Specimen / Site	Accession	Reported
[REDACTED, J.] Patient ID 44021 8 M	Cerebellar vermis mass, subtotal resection.	GA-2026-09102	2026-03-19

TIER I — FINDINGS WITH STRONG CLINICAL SIGNIFICANCE

Gene / Marker	Alteration	Variant Class	Functional Impact	Clinical Note
PTCH1	p.R691*	SNV	Pathogenic	Tier I
rna_signature	SHH-pathway transcriptional signature (centroid match: MB, SHH-activated, subgroup 3)	Signature	Activating	Subgroup-defining

BIOMARKER STATUS

Biomarker	Result
Microsatellite Instability (MSI)	MSS
Tumor Mutational Burden (TMB)	1.1 mut/Mb (cutoff for High: ≥10)
Methylation class (DKFZ classifier v12.5)	MB, SHH-activated (subgroup 3) — calibrated score 0.94

VARIANT DETAIL (TIER I – III)

*Variants below tiered per AMP/ASCO/CAP 2017. Tier IV (benign / likely benign / artifactual) variants are not displayed in this report; see CSV deliverable for full list.**

Gene	HGVSc	HGVSp	VAF	Tier / Class
PTCH1	c.2071C>T	p.R691*	0.49	Pathogenic — Tier I
TP53	wild-type	wild-type		Wild-type — confirmed by Sanger
SUFU	wild-type	wild-type		Wild-type

Copy number alterations:

Gene	Copy Number	Status
MYC	2	no amplification
MYCN	2	no amplification
GLI2	2	no amplification

Structural variants / expression signatures:

Kind	Description
rna_signature	SHH-pathway transcriptional signature (centroid match: MB, SHH-activated, subgroup 3)

METHYLATION CLASSIFIER

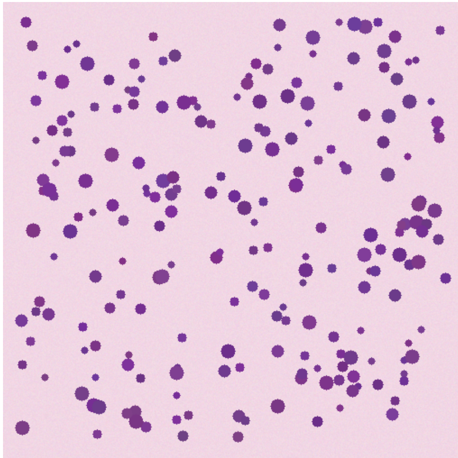
Methylation class (DKFZ classifier v12.5)
MB, SHH-activated (subgroup 3) — calibrated score 0.94

ANCILLARY HISTOLOGY / IHC SUMMARY

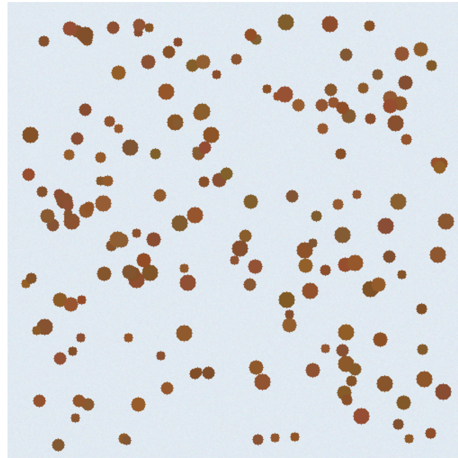
Morphology (per submitted slides): Densely cellular small round blue cell tumor with classic medulloblastoma morphology. Tumor cells exhibit hyperchromatic nuclei, scant cytoplasm, and brisk mitotic activity (>15/10 HPF in hotspots). Homer Wright rosettes identified focally. No large-cell or anaplastic features. No nodularity. Adjacent cerebellar parenchyma shows reactive gliosis.

Marker	Result
Synaptophysin	Diffuse positive
NeuN	Focal positive
INI1 (SMARCB1)	Retained nuclear expression
β-catenin	Cytoplasmic only (no nuclear translocation)
GAB1	Cytoplasmic, diffuse positive
YAP1	Nuclear and cytoplasmic positive
p53	Wild-type pattern (weak, patchy nuclear)
Ki-67 (MIB-1)	~45% in hotspots

REPRESENTATIVE IMAGES



H&E; ×20 — small round blue cell tumor



GAB1 IHC ×20 — cytoplasmic positivity

INTERPRETATION

Tier I: PTCH1 p.R691* pathogenic variant + SHH transcriptional signature + DKFZ methylation class MB-SHH (calibrated 0.94) establish SHH-activated subgroup. TP53 wild-type (Sanger confirmed) — places case in favorable SHH-WT category vs. high-risk SHH-mut. No MYC/MYCN amplification. INI1 retained excludes AT/RT. Suggest craniospinal staging and risk-adapted therapy per current pediatric MB protocol.

METHODOLOGY & LIMITATIONS

DNA / RNA extracted from FFPE tissue. Hybrid-capture next-generation sequencing of 612 cancer-associated genes; mean target coverage ≥250x. RNA expression signatures derived via medulloblastoma centroid classifier (MB-class v2.1). Methylation by Illumina EPIC v2.0 array, processed by the DKFZ classifier v12.5. Variant tiering per AMP/ASCO/CAP 2017. Tumor purity estimate: 71% (range 60-80%). Not reportable for variants below 5% VAF in low-purity samples.

Electronically signed: Dr. L. Chen, MD, FCAP — 2026-03-19
Co-signed: Dr. R. Park, PhD (Molecular Director)

ADDENDUM (2026-03-26)

Methylation classifier re-run after pathologist request: calibrated score confirmed at 0.94 (MB-SHH-3). Reproducibility OK. No change to molecular interpretation.

*Tier IV variants and assay performance metrics available upon request; see report deliverable bundle, file tier4_vus.csv.